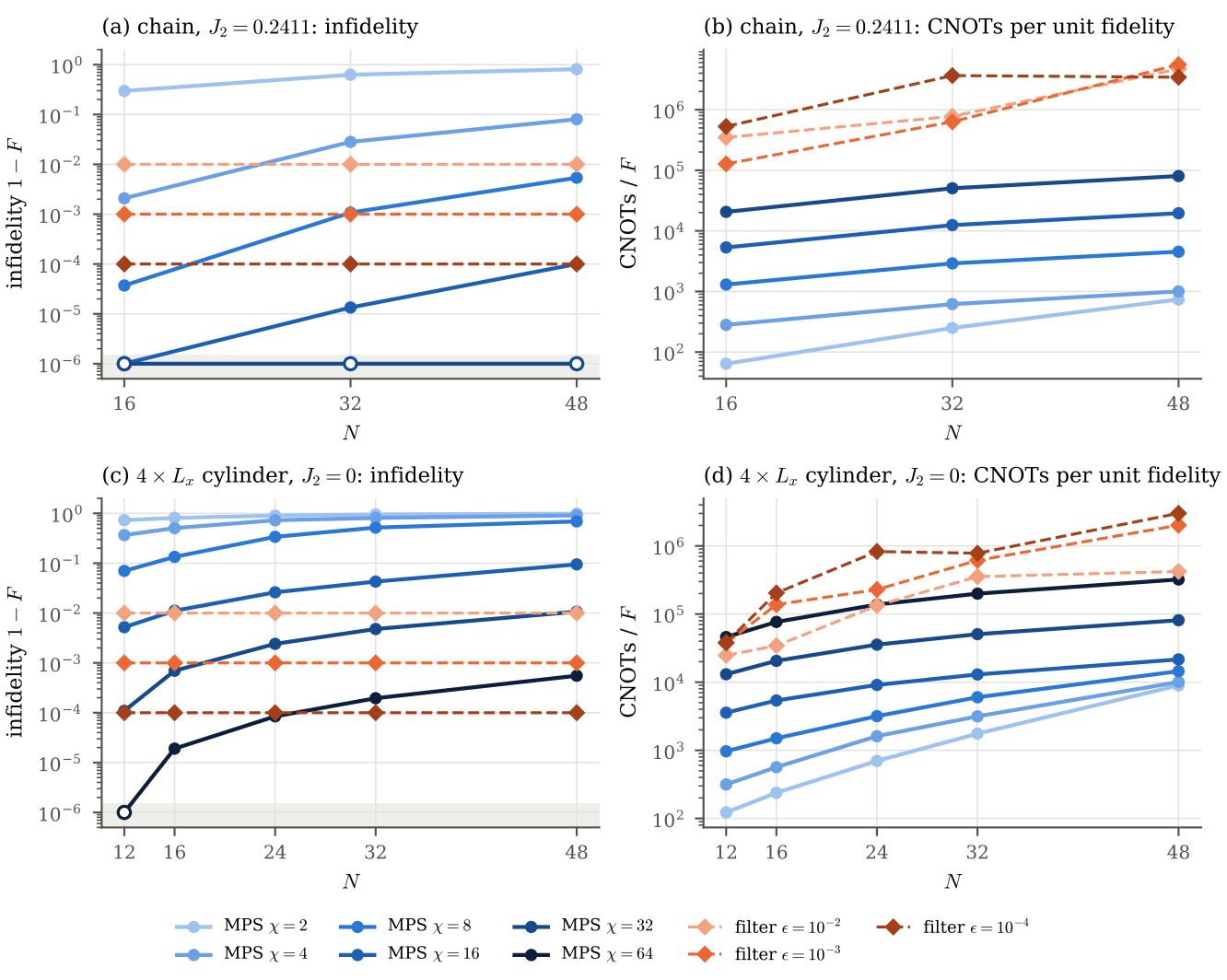
(a) chain, $J_2 = 0.2411$: infidelity
infidelity $1 - F$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
16
32
48
N
(b) chain, $J_2 = 0.2411$: CNOTs per unit fidelity
CNOTs / F
10^{6}
10^{5}
10^{4}
10^{3}
10^{2}
16
32
48
N
(c) $4 \times L_x$ cylinder, $J_2 = 0$: infidelity
infidelity $1 - F$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
12
16
24
32
48
N
(d) $4 \times L_x$ cylinder, $J_2 = 0$: CNOTs per unit fidelity
CNOTs / F
10^{6}
10^{5}
10^{4}
10^{3}
10^{2}
12
16
24
32
48
N
MPS $\chi = 2$
MPS $\chi = 4$
MPS $\chi = 8$
MPS $\chi = 16$
MPS $\chi = 32$
MPS $\chi = 64$
filter $\epsilon = 10^{-2}$
filter $\epsilon = 10^{-3}$
filter $\epsilon = 10^{-4}$